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PROJECT AND TEAM INFORMATION

Project Title

FoodRescue — Smart Food Rescue and Redistribution System

Team Name: *Team #*	*Avengers*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Dutta, Aditya – 2510373905* *2510373905@geu.ac.in* 
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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

FoodRescue is a project that allows food to go from places like restaurants, hostels, homes and events to NGOs and people who need it. It tracks donors NGOs' food donation needs volunteers and deliveries. It also ranks donations on the basis of time remaining till the the quantity needed and the distance expires and hashes donations to make records easier to find and uses a priority queue and Dijkstras algorithm to plan delivery routes. The program is written in C++ code. It will also build a web interface by HTML CSS and JavaScript. FoodRescue saves all records in CSV or text files.

Updated Project Approach and Architecture

Member 1. Donor & Food Donation Module
Classes: Donor, Food Donation
Include functions like adding donations updating donations deleting donations
Handle expiry dates and the availability of food
Use hashing to search among donors and donations
Member 2. NGO & Food Requirement Module
Classes: NGO, Food Requirement
NGO Login and Registration
Handle the requirements of food
Use a queue for the pending food requests
Match donations with the food that is requested
Member 3. Volunteer & Delivery Module
Classes: Volunteer, Delivery
Allocate volunteers to deliveries
Track the status of each delivery
Use queue or priority queue for deliveries
Keep the history of deliveries
Member 4. Core DSA & User Interface
Implement priority queue, for food and expiring food
Develop searching and sorting algorithms
Integrate the whole system together
Provide a menu-driven interface
Do testing and create the final document

Tasks Completed

Task Completed	Team Member
Created the donor class structure and added operations for working with donor details.	Raghav Pandey
Prepared the NGO class structures for storing managing NGO Information.	Vishwajeet Singh
Prepared classes that are used to store and manage food delivery details.	Utsav Bajoria
Worked on matching food donations with requirements and also arranging urgent donations with the help of priority-based logic.	Aditya Dutta

Challenges/Roadblocks

The difficulty lies in working out which donation should be given to which NGO, and the system must take into account expiry time, quantity and distance rather than focusing on a single factor. We intend to keep the donation and requirement records up to date and to use a priority score or a priority queue so that urgent food can be dealt with first. It is also necessary to verify the expiry dates and availability in order that food which cannot be distributed is not proposed for distribution. For route planning, the initial version will make use of a previously defined graph of locations and roads and will apply Dijkstra's algorithm to calculate a route. Another thing to do is to ensure that each team member's module functions correctly in conjunction with the others and that the records are not lost or duplicated. We will carry out module-level checks and tests using sample records to identify any problems. Live GPS, live maps and food-safety certification are not part of the current project scope.

Tasks Pending

Task Pending	Team Member (to complete the task)
Add hashing so donor and food donation records can be found quickly.	Raghav Pandey
Use a queue to keep track of food requests that are waiting to be processed.	Vishwajeet Singh
Add a priority queue to arrange food deliveries according to their priority.	Utsav Bajoria
Connect the front end with the application modules so users can interact with the system.	Aditya Dutta

Project Outcome/Deliverables

The objective is to develop a functional FoodRescue programme which enables individuals or organisations with surplus food to be linked with NGOs that require it. The intended features will cover donor, NGO, donation, requirement, volunteer and delivery records. The programme is also supposed to identify urgent donations, match donations with the needs of NGOs, determine routes by means of a graph and Dijkstra's algorithm, keep track of deliveries and store the records in CSV or text files. A web interface based on HTML, CSS and JavaScript is also planned. For the final submission, the C++ code, project documentation, test results and a demonstration are to be included.

Progress Overview

In addition to providing an overview of the problem and identifying system user classes, the proposal discusses the essential classes, planned data structure and algorithm usage, method of data storage, and projected outputs. However, it does not provide a specific percentage of work completed so far. Hence no exact figure is presented here in terms of project preparation. Among the remaining processes to be performed are linking together the different modules, verifying active and expired donations, developing the donation matching and route planning systems, testing the storage system for done files, and connecting the interface and the primary system. This section is also required to be updated with the latest codes and test results before submission.

Codebase Information

Repository link: https://github.com/duttta11/FOOD-RESCUE

Branch: Main.

Files/commits to note: DONOR.h, NGO_FINAL.h, FoodDonation DeliverySystem.h, PriorityQueue.h and UrgencyManager.h.

The main program is being developed in C++ using OOP and DSA. HTML, CSS and JavaScript are planned for the web interface. Check that the repository link, branch name and listed files are current before submitting the report.

Testing and Validation Status

Test Type	**Status (Pass/Fail)**	**Notes**
Donor and donation record operations	Pass	Test add update delete and lookup operations.
Expiry and availability validation	Pass	Confirm expired or unavailable food.
Matching NGO and donor.	Pass	Test priority ordering and matching algorithm.

Deliverables Progress

The project has completed the basic aspects, such as statement, proposal, and architecture design. In addition, the report mentions that the progress regarding the implementation of the donor and food donation module of the NGO and requirement-volunteer modules is not available. The plan incorporates priority-based matching and graph-based route planning but their implementation is yet to be obtained. Furthermore, the project includes delivery tracking, CSV/text file storage and web interface. The project progress needs to be monitored in relation to the organization of testing, final documentation, and demonstration.